

Name: _____

Date: _____

QUADRATICS, (FACTORISING QUADRATICS)

LO: I can factorise quadratic expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$.

Factorising quadratics

Factorising a quadratic means writing it as a product of two brackets. For $x^2 + bx + c$, find two numbers that **multiply to c** and **add to b**.

Strategy: List factor pairs of c. Find the pair that sums to b. Write in two brackets.

Quick examples

●	$x^2 + 7x + 12 = (x + 3)(x + 4)$	$3 \times 4 = 12, 3 + 4 = 7$
●	$x^2 - 5x + 6 = (x - 2)(x - 3)$	$-2 \times -3 = 6, -2 + -3 = -5$
●	$x^2 + x - 12 = (x + 4)(x - 3)$	$4 \times -3 = -12, 4 + -3 = 1$
●	$x^2 + 7x + 12 = (x + 6)(x + 2)$	$6 + 2 = 8, \text{ not } 7$
●	$x^2 - 5x + 6 = (x - 6)(x + 1)$	$-6 \times 1 = -6, \text{ not } +6$

Key idea

Find two numbers that multiply to c and add to b, then write
 $x^2 + bx + c = (x + m)(x + n)$.

$$x^2 + bx + c = (x + \text{m})(x + \text{n}) \text{ where}$$

$$mn = c \text{ and } m + n = b$$

Product = c, Sum = b

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - both positive

Factorise $x^2 + 8x + 15$

Need two numbers: multiply to 15,
add to 8

Factor pairs of 15: $1 \times 15, 3 \times 5$

$3 + 5 = 8$, so $(x + 3)(x + 5)$

$$(x + 3)(x + 5)$$

Both numbers positive because b and c are both positive.

Example 2 - both negative

Factorise $x^2 - 7x + 12$

Need: multiply to +12, add to -7
Both must be negative: $-3 \times -4 = 12$,

$-3 + -4 = -7$

$= (x - 3)(x - 4)$

$$(x - 3)(x - 4)$$

c positive but b negative means both numbers are negative.

Example 3 - mixed signs

Factorise $x^2 + 2x - 15$

Need: multiply to -15, add to +2

Try: $5 \times (-3) = -15, 5 + (-3) = 2$

$= (x + 5)(x - 3)$

$$(x + 5)(x - 3)$$

c negative means one number positive, one negative.

Example 4 - coefficient of x^2 not 1

Factorise $2x^2 + 7x + 3$

Multiply $a \times c = 2 \times 3 = 6$. Find pair:
product 6, sum 7

6 and 1: split middle term: $2x^2 + 6x$
 $+ x + 3$

Factorise in pairs: $2x(x+3) + 1(x+3) =$
 $(2x+1)(x+3)$

$$(2x + 1)(x + 3)$$

Use the grouping method when $a \neq 1$.

YOUR TURN – PRACTICE (deliberate practice)

1. Both terms positive

Factorise each quadratic.

- a) $x^2 + 5x + 6$
- b) $x^2 + 9x + 20$
- c) $x^2 + 11x + 30$
- d) $x^2 + 7x + 10$

2. Negative b, positive c

Factorise each quadratic.

- a) $x^2 - 5x + 6$
- b) $x^2 - 8x + 15$
- c) $x^2 - 9x + 14$
- d) $x^2 - 11x + 18$

3. Negative c

Factorise each quadratic.

- a) $x^2 + 3x - 10$
- b) $x^2 - 2x - 15$
- c) $x^2 + x - 20$
- d) $x^2 - x - 12$

4. Leading coefficient $\neq 1$

Factorise using the grouping method.

- a) $2x^2 + 5x + 3$
- b) $3x^2 + 10x + 3$
- c) $2x^2 + 7x + 6$
- d) $5x^2 + 11x + 2$

5. Mixed factorising

Factorise each expression fully.

- a) $x^2 + 4x - 21$
- b) $x^2 - 10x + 25$
- c) $x^2 - 49$
- d) $3x^2 - 5x - 2$
- e) $6x^2 + x - 2$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $x^2 + 7x + 12 = (x + 3)(x + 4)$ ☐

Correct answer: _____

Reason: _____

b) $x^2 - 5x + 6 = (x - 6)(x + 1)$ ☐

Correct answer: _____

Reason: _____

c) If c is negative, the signs in the brackets are different ☐

Correct answer: _____

Reason: _____

d) $x^2 + 7x + 12 = (x + 6)(x + 2)$ ☐

Correct answer: _____

Reason: _____

e) $x^2 - 9 = (x + 3)(x - 3)$ uses difference of two squares ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Factorise $6x^2 + 17x + 12$ completely. Verify your answer by expanding. Then solve $6x^2 + 17x + 12 = 0$.

Expression: _____

Simplified: _____

b) The area of a rectangle is $x^2 + 9x + 20$. Write down two possible expressions for the length and width. If the perimeter is **38 cm**, find x and the dimensions of the rectangle.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

